LESSON 9.3
FINDING AND INTERPRETING THE MEAN

LESSON INTRODUCTION

MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.

LEARNING TARGETS

- I can find and interpret the mean of a data set.

BENCHMARK COHERENCE

BUILDING ON

MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.

WORKING TOWARD

MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

ESSENTIAL QUESTIONS

- How can you find the mean of a data set?
- What does the mean tell you about the data set?
- What are some real-world examples of when mean can be useful?

LESSON NARRATIVE

Students will use data in a real-world context to find and interpret the mean of a data set. Students will discover how to evenly distribute a certain number of items into a predetermined number of groups.

COMPONENT SUMMARY

9.3.1 WARM-UP *(~5 MIN)*

Compare lengths of lines

9.3.2 EXPLORATION *(10 MIN)*

Discover how to redistribute groups of a square evenly

9.3.3 GUIDED INSTRUCTION *(10-15 MIN)*

Find the average (mean) of a given set of numbers

9.3.4 TRY IT *(~10 MIN)*

Practice finding the mean of a data set

9.3.5 YOUR TURN! *(~10 MIN)*

Prove understanding of finding the mean of data

9.3.6 WRAP-UP *(~5 MIN.)*

Define mean with words and visually

RESOURCES

GLOSSARY

Key Terms:

- Mean
- Measures of center

Additional Terms: N/A

MATERIALS

- (optional) Calculator
- (optional) Manipulatives
- (optional) Paper
- (optional) Sticky notes

ADDITIONAL PREPARATION

- If using sticky notes for the 9.3.6 Wrap-Up, consider creating an area on the wall before the lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may try to find the mean of categorical data.
- Students may forget to divide by the number of data values after finding their sum to determine the mean.
- Students may call the mean "median."
- Students may think the mean must be one of the values in the data set.

THINGS TO CONSIDER

- Some students may still struggle with basic division. Consider providing a chart or a calculator so that this is not the focus.
- Encourage students to notice how the mean may not be a number in the data set.
- Consider providing students with extra paper to solve and give more room to think.
- Encourage students to show their work on every problem to avoid basic addition mistakes.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

For 9.3.4 Try It, engage students in a Notice and Wonder routine before interpreting each data set. This provides students with the opportunity to begin to analyze the data and determine patterns.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Data throughout this lesson is presented through real-world contexts. Encourage students to connect the content of the lesson, particularly finding mean, to their everyday lives by considering when knowing an average might be helpful. This will build investment in the mathematics.

DIFFERENTIATE INSTRUCTION

Notice the opportunities to use manipulatives and visuals during 9.3.2 Exploration and 9.3.5 Your Turn!. Visual entry points are available through the display of data in graphs and charts. Encouraging the use of manipulatives in these sections will provide an additional kinesthetic entry point for learners to engage with the content. This is particularly helpful as students use manipulatives to create equal groups and compare this to the algorithm they may have used to find the mean. This connection deepens understanding of how to find the mean and can help students understand the context, particularly when determining the number needed to distribute evenly amongst groups. Students will see how certain data, such as animals or humans, cannot be split into fraction or decimals.

9.3.1 WARM-UP: STRIPES

Component Overview: *In this Warm-Up, students will compare stripes on four groups of cubes. Students will discuss the similarities and differences between the sizes.*

DIFFERENTIATE INSTRUCTION

For those who struggle with spatial reasoning, consider allowing students to cut out each colored strip to compare them more easily. Consider having students imagine how triangles could make up each line to also get a better visual on how long each is.

9.3.1 Warm-Up: Stripes

A picture of four sets of 16 cubes is shown.

1. Which set of cubes has a set of stripes that covers the most area?

   The set in the bottom left corner with the green stripes is the set with stripes that cover the most area.

2. Explain how you determined your answer.

   Answers may vary. It has the most stripes, so they logically cover the most area.

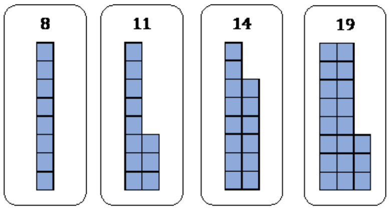
3. Which set of cubes has a set of stripes that covers the least area?

   The set in the top right corner with the blue stripes is the set with stripes that cover the least area.

4. Explain how you determined your answer.

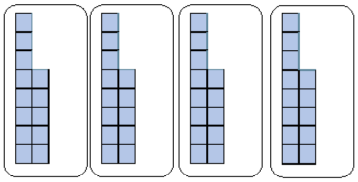
   Answers may vary. If you put the blue stripes together, they will still be shorter than the one red stripe.

9.3.2 Exploration: Sharing Squares



1. **Estimate the number of squares that will be in each group if they were distributed evenly between the four groups.**
   13

2. **Use the diagram to redistribute the squares so that each of the four groups has the same number of squares.**



3. **Compare diagrams with your partner. Discuss how you each determined the number of squares that belong in each group. Summarize your discussion.**
   Answers will vary. Add all the blocks together and then divide by the number of cards, 4. I moved the blocks around from the columns with 14 and 19 so that the number in each became more even.

9.3.2 EXPLORATION: SHARING SQUARES

Component Overview: *In this Exploration, students evenly distribute squares into groupings. Students translate this into working with adding and dividing numbers to find the mean.*

DIFFERENTIATE INSTRUCTION

Consider providing square manipulatives for students to help understand the best way to distribute the squares evenly into four parts. This will provide a kinesthetic entry point.

ENRICHMENT

Students may struggle to explain how they determine the number of squares that belong in each group. To provide further guidance for discussion, ask students to consider the following questions:

- What did you do first to consider how to divide the squares?
- What does it mean to distribute evenly? Will each group have the same number? Why?
- How does the number of groups impact your answer?

9.3.3 GUIDED INSTRUCTION: MEAN OF DATA

Component Overview: *In this Guided Instruction, students will find the mean of a data set using the given information. This page requires a heavy amount of reading, which may need to be broken down for struggling readers and ELLs. This section can be used as a guide for notetaking.*

TEACHER MOVES

Turn and Talk

This section is intentionally scaffolded to build conceptual understanding of what mean is and why it is used. Read through the definition of mean with the class, then have students engage in a Turn and Talk routine to restate the definition in their own words. This will ensure understanding over rote memorization.

DIFFERENTIATE INSTRUCTION

Consider providing a graphic organizer to frame student thinking and help set up the proper way to find the mean of a set of numbers. Students may struggle with seeing numbers vertically and may need to view them horizontally.

ENRICHMENT

Students may struggle to interpret the mean in question 2c. To further scaffold, ask questions like:

- What is the mean percentage of adults who use Instagram?
- Is that more or less than half? What does this say about the number of adults using Instagram?

TEACHER MOVES

Consider allowing students to turn this statement into a Madlib-style fill-in-the-blank to describe data in later problems.

9.3.3 Guided Instruction: Mean of Data

By redistributing the squares in the diagram, the ***mean*** of the data set was found.

The ***mean*** is the arithmetic average of a set of numbers. It is a measure of central tendency.

Data in a numerical set is summarized using different measures that are interpreted to describe the data. One such measure used to describe quantitative data sets is the mean.

To calculate the mean, add all the values in the data set and then divide by the total number of data values.

1. A marine biologist is studying turtles and wants to calculate the average length of time a Red-Eared Slider, a popular breed of turtle, can remain underwater. The time, in minutes, that ten Red-Eared Sliders remain underwater are 75, 300, 225, 400, 250, 195, 320, 45, 165, 410.
   a. To calculate the mean time spent underwater, start by finding the *sum* of the data values.
   The sum is 2,385.
   b. Then divide the sum by 10, because there are ten turtles.
   c. In this sample, Red-Eared Slider turtles remain underwater for an average of 238.5 minutes.

Measures of center are numerical values used to describe the overall clustering of data in a set, or the overall central value of a set of data. The three most common measures of central tendency are the mean, median, and mode.

2. Data from the Pew Research Center on the percentage of adults in the U.S. who use Instagram is shown in the table.

Percentage of U.S. Adults Using Instagram	
Year	**Percentage**
2012	13%
2013	17%
2014	26%
2015	29%
2016	34%

   a. Describe, in words, how to find the mean percentage of adults in the U.S. who use Instagram.
   Step 1: *Find the sum of the percentages.*
   Step 2: *Divide by 5, the number of years we have percentages for.*
   b. Find the mean percentage of adults in the U.S. who use Instagram. Think about what type of number the answer should be. 23.8%
   c. Interpret the mean based on the data. Specifically address what the value of the mean states about the number of adults who use Instagram.
   Between 2012 and 2016, about 23.8% of U.S. adults used Instagram.
   d. Explain why the mean does not give information about the percentage of adults in the U.S. using Instagram today.
   Answers will vary. The data only goes through 2016. Since the percentage has increased every year, it is likely that the mean would be higher than 23.8%.

The ***mean*** of a data set is one way to describe the center of the data set, by generalizing the data. In other words, it describes a typical value of the data set.

   e. Complete the statement to interpret the mean of the data set from the Pew Research Center.
   Typically, about 23.8% of adults in the U.S. used Instagram during the years 2012-2016.

9.3.4 Try It: Finding and Interpreting the Mean

1. The heights of seven NBA players are listed in the table.

Player	Height (in inches)
LeBron James	81
Stephen Curry	75
Kevin Durant	82
Kobe Bryant	78
Michael Jordan	78
James Harden	77
Shaquille O'Neal	85

a. What is the mean height, in inches, of the NBA players sampled? Round to the nearest tenth.
79.4 inches

b. Interpret the mean height for the seven players.
Answers will vary. The average height of these players is 79.4 inches.

c. Determine whether the mean should be used to make a generalization about all NBA players. Justify your answer.
Answers will vary. No, you cannot use a small sample to make a generalization about an entire group.

2. Below is a list of the number of home runs hit in the top 10 most home-run friendly ballparks during the 2012 Major League Baseball season.

231	202
230	199
229	175
225	174
218	158

Circle the value from the data set that is closest to the mean number of home runs per ballpark. Show your work.
204.1

9.3.4 TRY IT: FINDING AND INTERPRETING THE MEAN

Component Overview: *In this Try It, students work independently or with a partner to find and interpret the mean height of seven NBA players. Students find the mean number of homeruns hit and determine the value in the data set that is closest to the mean.*

TEACHER MOVES

Notice and Wonder

Engage students in a Notice and Wonder routine before interpreting each data set. This provides students with the opportunity to begin to analyze the data and determine patterns. Circulate as students share their findings with a partner, then bring the class together to draw attention to particularly helpful notices and wonderings like:

- I notice most of the NBA players' heights are similarly high.
- I wonder if this data is representative of all NBA players.
- I notice there is a large variation in the homerun data.

9.3.5 YOUR TURN!

Component Overview: *Students work independently to find and interpret the mean of data sets presented in tables. Students then model how to distribute data equally between groups and determine the number of kittens necessary to evenly distribute between a number of groups.*

TEACHER MOVES

MA.K12.MTR.7.1: Mathematics in Real-World Contexts

Students may struggle to interpret the data presented in the table if they do not consider the real-world context. Remind students of the differences between Florida and other states and ask them to consider why people visit Florida to help them make interpretations.

DIFFERENTIATE INSTRUCTION

Consider allowing students to use manipulatives to visualize how to evenly distribute the cats. Allow students to either place the cats into boxes like the diagram shown and redistribute or take out the total number of cats and place an equal number in each box. This provides a visual and kinesthetic entry point.

9.3.5 Your Turn!

1. **Statistics on the total number of visitors to Florida are tracked by Visit Florida Research. The table shows the number of visitors, in millions, for each quarter for the year 2019.**

Visitors to Florida (in millions)	
Quarter	**Total Visitors**
Q1 (January-March)	35.79
Q2 (April-June)	32.41
Q3 (July-September)	32.40
Q4 (October-December)	30.81

a. **What is the mean number of people, in millions, who visited Florida per quarter in 2019? Round your answer to the hundredths place.**
32.85 million visitors

b. **What is the mean number of people, in millions, who visited Florida per month in 2019? Round your answer to the hundredths place.**
10.95 million visitors

c. **Interpret the mean number of visitors per quarter in terms of the data set.**
Answers may vary. The average number of visitors to Florida per quarter in 2019 was 32.85 million.

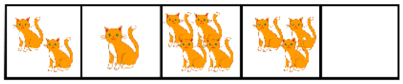
2. **For one week, Diego kept track of how long it took for his dad to drive him to school each morning.**

Diego's Travel Time to School (in minutes)	
Day	**Travel Time**
Monday	22
Tuesday	18
Wednesday	11
Thursday	8
Friday	11

a. **What is the mean travel time for Diego to get to school?**
14 minutes

b. **Interpret the mean in terms of the data set.**
Answers will vary. On average, it took Diego and his dad 14 minutes to drive to school that week.

3. **An animal shelter has 5 crates available to hold 10 kittens. The manager of the shelter wants the kittens to be distributed equally among the crates.**



a. **How many kittens will end up in each crate?**
2

b. **If the shelter receives 2 more kittens, how many kittens must be placed in each crate so they are equally distributed?**
There is no way to distribute the 12 kittens evenly among the 5 crates since a kitten can't be split. The 12 kittens could be split among 4 crates with 3 kittens in each crate for an even distribution.

c. **The shelter wants to keep kittens equally distributed and use all five crates. Circle the number of kittens that allows the shelter to equally distribute the kittens among five crates.**

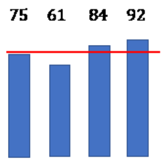
12	15	6	18	20	23	9

9.3.6 Wrap-Up: Cool Down

1. Explain what the **mean** of a data set represents.

   Mean is the average of a set of numbers.

2. What is the average height of the bars? Draw a horizontal line across the bars to show the height of the bars if you redistributed the values so that all of the bars showed the same height.



The average height of the bars is 78.

9.3.6 WRAP-UP: COOL DOWN

Component Overview: *In this Wrap-Up, students will show their understanding of the concept of mean. Students define mean and then find the mean of data presented visually.*

DIFFERENTIATE INSTRUCTION

Consider using this Wrap-Up as a formative assessment for future lessons to inform small group instruction or partner pairs for activities.